

(ECBS5155) Coding 3: Introduction to R

Academic year: 2025

Background and overall aim

This is a hands-on and tech-focused course to get you up to speed using the R ecosystem (installing R, installing packages from CRAN and GitHub, manuals and vignettes, RStudio IDE, running R remotely) and R programming language for data transformations, data visualization, modeling, exploratory data analysis, and publishing HTML and PDF reports.

Course prerequisites

Coding 1: Data Management and Analysis with Python

Laptop with R, RStudio and git installed is required in the class and for the take-home assignments. Installation steps will be shared on GitHub.

Waiting list handling

Students outside of the MS in Business Analytics program will be placed on the waiting list and admitted on first-come-first-serve bases, provided that the pre-requisites are met.

Learning outcomes

By the end of the course, you will be able to use R to read and write data in different formats, do data transformations (e.g. filtering, aggregation, reshaping between long and wide tables, creating new variables), data visualizations (e.g. scatterplots, line plots, histograms, barcharts, boxplots), train basic models (e.g. linear regression, multidimensional scaling, tree-based methods), and generate reports in various formats (e.g. HTML and PDF).

Learning activities and teaching methods

The course will involve coding sessions with demos and exercises. Learning objectives will be achieved through actively taking part in the in-class exercises and solving homework and the final take-home assignment.

Assessment

40% homework and 60% final project.

Students shall not miss more than 1 day of classes, failing to do so will yield an administrative fail grade. To pass, students will need to get at least 50% of the homework AND at least 50% of the final project. Grading will be based on the total score out of 100, in line with CEU's standard grading guidelines.

Course contents

Topic Outline:

1. R ecosystem (base R, packages, RStudio), data frames, basic plots, reading CSV files, intro to modeling
2. Advanced plotting with ggplot2
3. Data transformations using data.table
4. Modeling with base R, rpart, and randomForest
5. Publication-ready reports using R markdown, modeling

Reading list:

Class materials will be available on GitHub. Databases The CEU Library boasts a range of databases covering financial and company data, market and industry reports, global news and more. For a full list of databases visit the CEU Library.

- Refinitiv (Thomson Reuters) Eikon for Students + Datastream/Thomson ONE
 - Eikon: Platform used by finance practitioners including market traders to monitor and analyze financial information. Information, analytics and news on all major financial markets including real-time pricing data, financial research, global financial news and commentary, financial estimates, fundamentals analysis, visual analysis through charting. Import/export from Excel.
 - Datastream: Range of economic, securities and company financial data. Excel add-in.
 - Thomson ONE: Global overviews on 55,000 public companies, one million private companies. Reuters News, ownership, deals, private equity, key ratios, company filings, officers and directors. Investext analyst reports, active and historical research from 1,600 independent research firms, brokerages, investment banks.
- Standard & Poor's Capital IQ
 - Web and Excel-based platform combining deep global company information, credit ratings and research, and market research with powerful tools for risk assessments. Real-time and historical information on markets, industries, companies, transactions and people. Tearsheet data.
- Lexis Nexis Academic
 - Global database of news, business, legal and other sources. Full text of 350 newspapers, 300 magazines and journals, 600 newsletters. Wire services including Associated Press, Business Wire and PR Newswire. Company financial information, market research, industry reports.

Contact details

Short bio of instructor:

Gergely Daroczi - has a PhD in Sociology, 15 years of experience with R, founder of the Hungarian R meetup and main organizer of R conferences, authored a book on R and maintains a dozen of R packages, lived and worked in Hungary and USA at market research, fintech, adtech and healthtech companies as a data scientist and engineer both in individual contributor and management roles.

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Practical

Course level	7
Ac. year	2025-2026
Term	Spring term
Host Unit	Department of Economics
US Credits	1
ECTS Credits	2
Course code	ECBS5155
requires	ECBS5208 Coding 1: Introduction to Python
Offered for non-degree students	N